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CENTRE
NUMBER

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CANDIDATE
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CHEMISTRY

Grade 12

Paper 2

May 2016

2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet
 Electronic calculator

12CHEM/02

READ THESE INSTRUCTIONS FIRST

Write your Centre number and candidate number in the spaces at the top of the page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

Electronic calculators may be used.

You may lose marks if you do not show your working or if you do not use appropriate units.

A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

Answer all questions in English.

For Examiner's Use

1	
2	
3	
4	
5	
6	
7	
8	
9	
Total	

This document consists of **17** printed pages and **3** blank pages.

- 1 Ammonium nitrate, NH_4NO_3 , is a solid white powder at room temperature. It is a salt and can be made by reacting ammonia with a suitable acid.

For
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Use

- (a) Write a balanced chemical equation for this reaction.

..... [1]

- (b) The formula of the ammonium ion is NH_4^+ .

- (i) Draw a dots and crosses diagram of an ammonium ion.

[2]

- (ii) State the H-N-H bond angle in an ammonium ion.

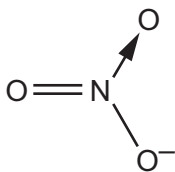
bond angle [1]

- (iii) The four N-H bonds in each ammonium ion are all covalent, but they did not all form in the same way.

Explain this statement as fully as you can. Refer to your dots and crosses diagram if it is helpful.

.....
.....
.....
..... [2]

(c) The formula of the nitrate ion is NO_3^- .



(i) Draw a dots and crosses diagram of a nitrate ion.

[2]

(ii) State the O-N-O bond angle in a nitrate ion.

bond angle [1]

(d) There are 42 electrons in each formula unit of ammonium nitrate. There are 42 electrons in each molecule of benzene. The melting point of benzene is lower than the melting point of ammonium nitrate.

Explain why there is a difference in melting point.

.....

.....

.....

.....

.....

..... [4]

[Total: 13]

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2 Compound **X** is a compound of carbon, hydrogen and oxygen.

- (a) Analysis of a sample of **X** shows that it is 54.55% carbon, 9.09% hydrogen and 36.36% oxygen by mass. When **X** is analysed using a mass spectrometer, the molecular ion peak has an m/z value of 88.

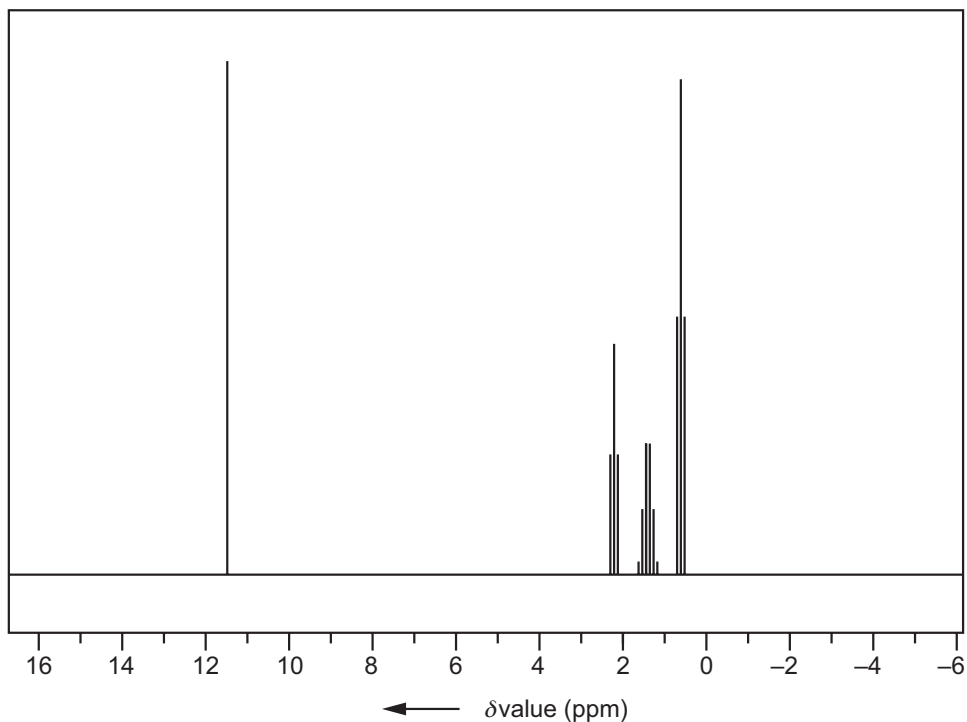
Use this information to show the molecular formula of **X** is $C_4H_8O_2$.
Show all your working clearly.

For
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Use

[3]

(b) When **X** is analysed by proton NMR spectroscopy, the spectrum below is obtained.

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(i) Explain why the peak at $\delta = 0.6$ is split into a triplet.

.....
 [1]

(ii) Deduce the structural formula of **X**. Use the proton NMR spectrum to explain your decision. Identify the protons responsible for each of the absorptions in the spectrum.

.....

 [5]

- (c) **X** has four structural isomers all with the same functional group. This functional group, which may be hydrolysed by hot NaOH(aq), is **different** to the functional group in **X**. Draw **and** name the structural formulae of any **two** of these isomers.

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isomer

name

isomer

name

[4]

[Total: 13]

3 Group 14 of the Periodic Table contains the elements carbon, silicon, germanium, tin and lead. These elements form compounds in two oxidation states. For example:

- carbon forms two oxides, CO and CO₂
- lead forms two oxides, PbO and PbO₂

(a) (i) Identify the oxidation state of carbon in CO and in CO₂.

..... [1]

(ii) Which oxidation state of carbon is more stable?

.....

Write a balanced chemical equation for a reaction involving CO and CO₂ in which the more stable oxidation state of carbon is formed from the less stable oxidation state.

..... [2]

(b) PbO₂ is an oxidising agent. Write a balanced chemical equation for the reaction that takes place when solid PbO₂ is added to an acidified solution of FeSO₄.

..... [2]

(c) When an aqueous solution containing an excess of Sn²⁺ ions is added to an acidified solution containing VO₃⁻ ions, the VO₃⁻ ions are reduced. Use $E^\ominus$ values to predict the final oxidation state of vanadium in the solution. Explain your answer fully.

.....

 [4]

- (d) How do the stabilities of the +2 and +4 oxidation states of the elements in Group 14 vary going down the group?

Use examples taken from at least **two** of parts (a), (b) and (c) of this question.

.....

.....

.....

.....

.....

..... [3]

[Total: 12]

4 Data needed for this question:

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$$\Delta H_f^\circ \text{ MgO} = -602 \text{ kJ mol}^{-1}$$

$$\Delta H_f^\circ \text{ NO}_2 = +34 \text{ kJ mol}^{-1}$$

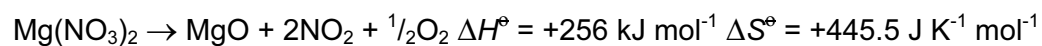
$$S^\circ \text{ O}_2 = +205 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$S^\circ \text{ MgO} = +27 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$S^\circ \text{ NO}_2 = +240 \text{ J K}^{-1} \text{ mol}^{-1}$$

ΔH_f° = standard enthalpy change of formation. S° = standard entropy

When anhydrous magnesium nitrate is heated it decomposes, giving magnesium oxide and a mixture of gases.



- (a) Use Hess' Law to calculate the standard enthalpy change of formation, ΔH_f° , of anhydrous magnesium nitrate, $\text{Mg(NO}_3)_2$.

..... kJ mol^{-1} [2]

- (b) Calculate the standard entropy, S° , of anhydrous magnesium nitrate, $\text{Mg(NO}_3)_2$.

..... $\text{J K}^{-1} \text{ mol}^{-1}$ [2]

- (c) Calculate the lowest temperature at which the decomposition of magnesium nitrate proceeds spontaneously. Give your answer to **three** significant figures.

temperature = [3]

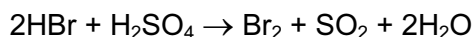
[Total: 7]

5 Sulfuric acid, H_2SO_4 , is an important industrial chemical.

- (a) (i) Sulfuric acid behaves as an acid in many of its reactions. Write a balanced chemical equation, including state symbols, for a reaction in which sulfuric acid behaves as an acid.

..... [1]

- (ii) Concentrated sulfuric acid acts as an oxidising agent in its reaction with hydrogen bromide.

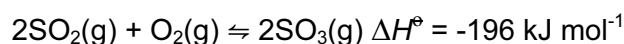


Identify the element that is oxidised and the element that is reduced in this reaction. Use oxidation numbers to justify your answer.

.....

 [2]

- (b) Sulfuric acid is made by the Contact process. The most important reaction in the Contact process is shown.



- (i) Name the catalyst that is used to speed up this reaction.

..... [1]

- (ii) This reaction is reversible. Predict, with reasons, the reaction conditions that will give the highest possible yield of SO_3 at equilibrium.

.....

 [3]

- (c) If SO_2 gas escapes into the environment, it can be responsible for an environmental problem.
 Name this problem and describe one consequence of it.

.....

 [2]

[Total: 9]

- 6 Methanoic acid, HCOOH , is a weak acid. When methanoic acid is dissolved in water, an equilibrium is established. This equilibrium involves H^+ ions and methanoate ions, HCOO^- .



- (a) Methanoic acid is an example of a Brønsted-Lowry acid. The methanoate ion is an example of a Lewis base. Explain these two statements by reference to the equation above.

.....

 [2]

- (b) What is the $\text{p}K_a$ of methanoic acid.

$\text{p}K_a$ [1]

- (c) If 100 cm^3 of $0.100 \text{ mol dm}^{-3}$ methanoic acid solution is mixed with 300 cm^3 of $0.100 \text{ mol dm}^{-3}$ sodium methanoate solution, 400 cm^3 of a *buffer solution* is formed.

- (i) Explain what is meant by a *buffer solution*, using at least two equations to show **how** it works as a buffer.

.....

 [4]

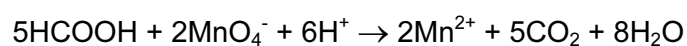
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- (ii) 0.730g of HCl gas is dissolved in the 400 cm³ of buffer solution. Calculate the pH of the solution formed when all of the HCl gas has reacted with the buffer.

For
Examiner's
Use

[5]

- (d) Methanoic acid can be oxidised by acidified potassium manganate(VII) solution.



50.00 cm³ of a solution of methanoic acid needs 17.25 cm³ of an acidified 0.0200 mol dm⁻³ solution of MnO₄⁻ for complete oxidation. Calculate the concentration of the methanoic acid solution.

[3]

[Total: 15]

- 7 Chloroethane, $\text{CH}_3\text{CH}_2\text{Cl}$, and ethanoyl chloride, CH_3COCl , will both react with water. The mechanism of both reactions is nucleophilic substitution.

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- (a) Explain the mechanism of the reaction between chloroethane and water.

You should:

- use a curly arrow to show the movement of an electron pair
- label any relevant dipoles
- show any relevant lone pairs of electrons
- include the formulae of the final products.

[4]

- (b) If aqueous silver nitrate, AgNO_3 , is added to chloroethane, a white precipitate forms very slowly. If aqueous silver nitrate is added to ethanoyl chloride a white precipitate forms immediately.

Name the substance seen as a white precipitate and suggest a reason why ethanoyl chloride is more reactive than chloroethane.

.....

 [2]

- (c) (i) Benzene can be used to make 1-phenylethanol, $\text{C}_6\text{H}_5\text{CH}(\text{OH})\text{CH}_3$, in a two-step process. Ethanoyl chloride is involved in the first step.

Suggest how this synthesis of 1-phenylethanol could be performed.
Your answer should include:

- any substances used as reagents or catalysts
- the structural formula of the compound formed in the first step.

[3]

- (ii) This synthesis will produce 1-phenylethanol as a mixture of two stereoisomers.
Draw the two stereoisomers of 1-phenylethanol.

[2]

[Total: 11]

- 8** Crude oil is an important source of hydrocarbons that are used as fuels and as a source of chemicals. These hydrocarbons include hexane and benzene.

(a) (i) Crude oil is processed to give mixtures called fractions. Name the process used.

..... [1]

(ii) Explain why this process separates the crude oil into fractions.

..... [1]

(b) Hexane can be catalytically dehydrogenated to give hex-3-ene. This is a decomposition reaction.

(i) Write a balanced chemical equation for this reaction.

..... [1]

(ii) Hex-3-ene exists as a mixture of two stereoisomers. Draw and name the two stereoisomers of hex-3-ene.

isomer name

isomer name [2]

(c) Hex-3-ene and benzene are both unsaturated but they have different chemical properties.

(i) Describe a chemical reaction that would give a different, visible, result with hex-3-ene and benzene. Include in your answer the observations that would be made.

.....
.....
..... [2]

- (ii) A molecule of hex-3-ene and a molecule of benzene both have π -bonding.
Draw two labelled diagrams to show the π -bonding in hex-3-ene and in benzene.

[2]

- (iii) Explain why the π -bonding in benzene gives additional stability to the molecule.

[1]

- (iv) State a piece of thermochemical evidence for this additional stability of benzene.

[3]

[Total: 13]

- 9 DNA is found in the nucleus of most animal cells. The sequence of bases in a section of a DNA molecule is a store of information. This information is used to produce the primary structure of a protein molecule. Protein molecules are polymers made from amino acids. Proteins are synthesised in the cytoplasm of animal cells.

(a) What is meant by the *primary structure* of a protein molecule?

..... [1]

(b) Name the **specific** type of molecule which is used to carry the information from the cell nucleus to the cell cytoplasm.

..... [1]

(c) The information in this molecule is read in groups of bases called codons. How many bases are there in a codon?

..... [1]

(d) If a non-functioning protein is made the organism may suffer from a disease.

Describe and name a change in a DNA molecule that could result in the production of a non- functioning protein.

.....
..... [1]

(e) A non-functioning protein may exist as a different shaped molecule compared with the functioning protein it has replaced.

Explain how a change in a protein may result in a different shaped protein molecule.

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.....
.....
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.....
.....
..... [3]

[Total: 7]

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